

Based on the given information, answer the following subquestions.

- (i) Write the fitted linear regression model for all the different models. [3 Marks]
- (ii) Write null and alternative hypotheses for the first model and check if we can reject the null hypothesis at 5% significance level, i.e., $\alpha = 0.05$. Also, what can you conclude about the relationship between the variables Hours of self study per week and Quiz marks. [3 Marks]
- (iii) Interpret R^2 for the first two models and explain which model will perform better. [2 Marks]

4. Consider a linear model as

$$y_i = \beta_1 x_i + \epsilon_i \quad ; \quad 1 \leq i \leq n$$

where, $\epsilon_i \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2)$.

- (i). Find the value of β_1 for which RSS (Residual Sum of Squares) will be minimum. Provide all the steps. [3 Marks]
- (ii). Find the value of $Var(\hat{\beta}_1)$. [2 Marks]
- (iii). Suppose $\epsilon_i \stackrel{\text{i.i.d.}}{\sim} N(\mu, \sigma^2)$ for the model $y_i = \beta_1 x_i + \epsilon_i \quad ; \quad 1 \leq i \leq n$. Find $bias(\hat{\beta}_1)$ [2 Marks]

5. Define Linear Zero Estimators. [2 Marks]

Options :

6406535903404. ✓ I have written answers on the answer sheets

6406535903405. ✗ Not applicable

Corporate Finance

Section Id :	640653121918
Section Number :	13
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	100
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes

Maximum Instruction Time : 0
Sub-Section Number : 1
Sub-Section Id : 640653318112
Question Shuffling Allowed : No

Question Number : 261 Question Id : 6406531815872 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : CORPORATE FINANCE (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE [TOP](#) FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406535903406. ✓ YES

6406535903407. ✗ NO

Sub-Section Number : 2
Sub-Section Id : 640653318113
Question Shuffling Allowed : Yes

Question Number : 262 Question Id : 6406531815873 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

According to the Expected Utility Hypothesis, if there are two possible states, 1 and 2, with probabilities π_1 and π_2 , and respective state-contingent consumptions are c_1 and c_2 , then the utility function is given by:

Options :

6406535903408. ✗ $U(c_1, c_2) = \pi_1 c_1 + \pi_2 c_2$

6406535903409. ✗ $U(c_1, c_2) = \pi_1 u(c_2) + \pi_2 u(c_1)$

6406535903410. ✗ $U(c_1, c_2) = u(\pi_1 c_1 + \pi_2 c_2)$

6406535903411. ✓ $U(c_1, c_2) = \pi_1 u(c_1) + \pi_2 u(c_2)$

Sub-Section Number : 3
Sub-Section Id : 640653318114
Question Shuffling Allowed : Yes

Question Number : 263 Question Id : 6406531815874 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Global Bank has total assets of 2,500 units, liabilities of 1,500 units, and capital of 1,000 units. What is the leverage ratio for Global Bank?

Options :

6406535903412. ✖ 0.4

6406535903413. ✖ 0.6

6406535903414. ✖ 1.67

6406535903415. ✔ 2.5

Sub-Section Number :

4

Sub-Section Id :

640653318115

Question Shuffling Allowed :

Yes

Question Number : 264 Question Id : 6406531815875 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

On a Mean-Standard Deviation diagram (Mean on the vertical axis and Standard deviation on horizontal axis) the left boundary of the feasible set represents the:

Options :

6406535903416. ✖ Efficient Frontier

6406535903417. ✖ Global Minimum Variance (GMV) portfolio

6406535903418. ✖ Capital Allocation Line (CAL)

6406535903419. ✔ Minimum-Variance Set

Sub-Section Number :

5

Sub-Section Id :

640653318116

Question Shuffling Allowed :

Yes

Question Number : 265 Question Id : 6406531815876 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A 1-year government bond offers a 5% rate of return. A lender, who is risk neutral, lends ₹50,000 to a friend. The lender believes the friend will pay back the promised amount with an 80% probability and will pay back nothing with a 20% probability. What is the default premium (in addition to the time premium) that the lender should ask from the friend?

Options :

6406535903420. ✖ 31.25%

6406535903421. ✔ 26.25%

6406535903422. ✖ 6.25%

6406535903423. ✖ 5.00%

Sub-Section Number : 6
Sub-Section Id : 640653318117
Question Shuffling Allowed : Yes

Question Number : 266 Question Id : 6406531815877 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

There are two assets: a risk-free asset with a 4% rate of return and a risky asset with an expected rate of return of 19% and a standard deviation of 10%. What is the slope of the Capital Allocation Line (CAL) of a portfolio consisting of these two assets?

Options :

6406535903424. ✖ 1.25

6406535903425. ✔ 1.5

6406535903426. ✖ 1.75

6406535903427. ✖ 2.0

Sub-Section Number : 7
Sub-Section Id : 640653318118
Question Shuffling Allowed : Yes

Question Number : 267 Question Id : 6406531815878 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A person's utility function is $U(r) = E(r) - \frac{1}{2}A\sigma^2$. If the person is indifferent between a risk-free asset (X) yielding 5% and a risky asset (Y) with $E(r) = 14\%$ and $\sigma = 12\%$, what is the coefficient of risk aversion (A)?

Options :

6406535903428. ✖ 10.0

6406535903429. ✔ 12.5

6406535903430. ✖ 15.0

6406535903431. ✖ 18.0

Sub-Section Number : 8
Sub-Section Id : 640653318119
Question Shuffling Allowed : Yes

Question Number : 268 Question Id : 6406531815879 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

An asset (A) has a standard deviation of 20%. The market portfolio (M) has a standard deviation of 5%. The correlation between the asset and the market is 0.2. What is the beta (β) of asset A?

Options :

6406535903432. ✖ 0.2

6406535903433. ✔ 0.8

6406535903434. ✖ 1.0

6406535903435. ✖ 3.2

Sub-Section Number :

9

Sub-Section Id :

640653318120

Question Shuffling Allowed :

Yes

Question Number : 269 Question Id : 6406531815880 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A two-asset portfolio has $\bar{r}_1 = 10\%$, $\bar{r}_2 = 30\%$. To achieve a target expected return of 15%, what weight must be invested in Asset 1?

Options :

6406535903436. ✖ 25%

6406535903437. ✖ 40%

6406535903438. ✔ 75%

6406535903439. ✖ 50%

Sub-Section Number :

10

Sub-Section Id :

640653318121

Question Shuffling Allowed :

Yes

Question Number : 270 Question Id : 6406531815881 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A two-asset portfolio consists of Asset X ($\sigma = 20\%$) and Asset Y ($\sigma = 10\%$). The correlation between the two assets is 0.5. If 60% is invested in X and 40% in Y, what is the portfolio variance (σ_p^2)?

Options :

6406535903440. ✔ 0.0208

6406535903441. ✖ 0.0300

6406535903442. ✖ 0.0484

6406535903443. ✖ 0.1442

Sub-Section Number :

11

Sub-Section Id :

640653318122

Question Shuffling Allowed :

Yes

Question Number : 271 Question Id : 6406531815882 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

An equally weighted portfolio is formed from 25 uncorrelated assets. Each asset has a standard deviation of 25%. What is the standard deviation of the portfolio?

Options :

6406535903444. ✖ 1%

6406535903445. ✔ 5%

6406535903446. ✖ 10%

6406535903447. ✖ 25%

Sub-Section Number :

12

Sub-Section Id :

640653318123

Question Shuffling Allowed :

Yes

Question Number : 272 Question Id : 6406531815883 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

What is the fair price of a bet that pays 20 with 20% probability, 50 with 50% probability, and 90 with 30% probability?

Options :

6406535903448. ✖ 36

6406535903449. ✔ 56

6406535903450. ✖ 66

6406535903451. ✖ 46

Sub-Section Number :

13

Sub-Section Id :

640653318124

Question Shuffling Allowed :

Yes

Question Number : 273 Question Id : 6406531815884 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A farm is worth 40 thousand under flood (20% probability) or 150 thousand in no flood situation (80% probability) next year. The farm was purchased for 100 thousand, financed by a loan of 80 thousand and equity of 20 thousand. The loan requires a promised repayment of 110 thousand. What is the expected payoff for the equity owner next year (in thousands)?

Options :

6406535903452. ✖ 0

6406535903453. ✖ 20

6406535903454. ✔ 32

6406535903455. ✖ 40

Sub-Section Number :

14

Sub-Section Id :

640653318125

Question Shuffling Allowed :

Yes

Question Number : 274 Question Id : 6406531815885 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

For a three-asset portfolio of uncorrelated assets, w_1, w_2, w_3 are the weights and σ^2 is the common (same for the three assets) variance. The portfolio variance is $\sigma_p^2 = w_1^2\sigma^2 + w_2^2\sigma^2 + w_3^2\sigma^2$. To find the Global Minimum Variance (GMV) portfolio, you would minimize σ_p^2 subject to:

Options :

6406535903456. ✖ $w_1 = w_2 = w_3$

6406535903457. ✔ $w_1 + w_2 + w_3 = 1$

6406535903458. ✖ $w_1 = w_2 = w_3 = 1$

6406535903459. ✖ $w_1 + w_2 + w_3 = 0$

Sub-Section Number :

15

Sub-Section Id :

640653318126

Question Shuffling Allowed :

Yes

Question Number : 275 Question Id : 6406531815886 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

An investor with mean-variance preferences $U = \mu - 20\sigma^2$ is comparing Asset A ($\mu = 0.175, \sigma^2 = 0.02$) and Asset B ($\mu = 0.15, \sigma^2 = 0.01$). Which asset would investor prefer?

Options :

6406535903460. ✖ Asset A

6406535903461. ✔ Asset B

6406535903462. ✖ Indifferent

6406535903463. ✖ Depends on initial wealth

Sub-Section Number :

16

Sub-Section Id :

640653318127

Question Shuffling Allowed :

Yes

Question Number : 276 Question Id : 6406531815887 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

The risk-free rate of return is 6% and the expected market return is 15%. If the beta (

β) of an asset is 1.2, then what is the expected return of the asset according to the Capital Asset Pricing Model (CAPM)?

Options :

6406535903464. ✓ 16.8%

6406535903465. ✗ 12.0%

6406535903466. ✗ 10.8%

6406535903467. ✗ 14.4%

Sub-Section Number :

17

Sub-Section Id :

640653318128

Question Shuffling Allowed :

Yes

Question Number : 277 Question Id : 6406531815888 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

An investor decides to short sell 100 shares of Tech Inc. The current share price is Rs. 300. At the end of 1 year, the share price drops to Rs. 275. How much profit does the investor make from exercising the short sale on these shares (Assuming there are no additional costs for this trade)

Options :

6406535903468. ✓ Rs. 2500

6406535903469. ✗ Rs. 2750

6406535903470. ✗ Rs. 3000

6406535903471. ✗ The investor incurs a loss.

Sub-Section Number :

18

Sub-Section Id :

640653318129

Question Shuffling Allowed :

No

Question Id : 6406531815889 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (278 to 279)

Question Label : Comprehension

A new vineyard is purchased for 700 thousand units. There is a 10% chance of a severe pest infestation in the next year. If there is an infestation, the total payoff of the vineyard will be 300 thousand units in the next year. Otherwise, the vineyard generates a payoff of 900 thousand units in the next year. The appropriate cost of capital is 15% per year. The purchase is financed with a loan of 500 thousand units and the remaining amount as equity.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 278 Question Id : 6406531815890 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

What would be the appropriate promised rate of return that the creditor of the loan would demand?

Options :

6406535903472. ✖ 33.33%

6406535903473. ✔ 21.11%

6406535903474. ✖ 30.88%

6406535903475. ✖ 25.00%

Question Number : 279 Question Id : 6406531815891 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

What is the expected payoff for the equity owner in the next year?

Options :

6406535903476. ✖ 133.33 thousand

6406535903477. ✖ 166.67 thousand

6406535903478. ✖ 188.89 thousand

6406535903479. ✔ 265.00 thousand

Sub-Section Number :

19

Sub-Section Id :

640653318130

Question Shuffling Allowed :

No

Question Id : 6406531815892 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (280 to 281)

Question Label : Comprehension

Anjali, an entrepreneur, has a utility function of the form $U(x) = \sqrt{x}$. If her new product is successful, she expects to earn 1,600 units. If it fails, she will earn 400 units. There is a 40% chance that her product will fail.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 280 Question Id : 6406531815893 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

What is Anjali's expected utility from the venture?

Options :

6406535903480. ✖ 40

6406535903481. ✖ 36

6406535903482. ✔ 32

6406535903483. ✖ 24

Question Number : 281 Question Id : 6406531815894 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Anjali can buy an insurance policy that pays her 1,200 units if the product fails. If she pays a price p for this policy, she would be sure to have an income of $1,600 - p$ regardless of success or failure. What would be the largest price (p) that Anjali would be willing to pay for such an insurance?

Options :

6406535903484. ✖ 256 units

6406535903485. ✖ 360 units

6406535903486. ✖ 484 units

6406535903487. ✔ 576 units

Sub-Section Number :

20

Sub-Section Id :

640653318131

Question Shuffling Allowed :

No

Question Id : 6406531815895 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (282 to 286)

Question Label : Comprehension

Suppose there are two assets, Asset 1 and Asset 2, with the following characteristics:

- Expected Rate of Return: $r_1 = 0.10$, $r_2 = 0.25$
- Standard Deviation: $\sigma_1 = 0.20$, $\sigma_2 = 0.40$
- Covariance: $\sigma_{12} = 0.03$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 282 Question Id : 6406531815896 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A portfolio is formed with 30% of wealth in Asset 1 and 70% of wealth in Asset 2. What would be the expected rate of return of this portfolio?

Options :

6406535903488. ✖ 0.195

6406535903489. ✔ 0.205

6406535903490. ✖ 0.220

6406535903491. ✖ 0.235

Question Number : 283 Question Id : 6406531815897 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

What would be the standard deviation of the rate of return of the portfolio described in previous question?

Options :

6406535903492. ✖ 0.396

6406535903493. ✖ 0.542

6406535903494. ✔ 0.308

6406535903495. ✖ 0.234

Question Number : 284 Question Id : 6406531815898 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Suppose you would like to construct a portfolio with an expected rate of return of 19%. What weights would you put on Asset 1 and Asset 2, respectively?

Options :

6406535903496. ✔ 40%, 60%

6406535903497. ✖ 50%, 50%

6406535903498. ✖ 60%, 40%

6406535903499. ✖ 70%, 30%

Question Number : 285 Question Id : 6406531815899 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

What are the weights of Asset 1 and Asset 2 in the minimum variance portfolio?

Options :

6406535903500. ✔ 93%, 7%

6406535903501. ✖ 83%, 17%

6406535903502. ✖ 73%, 27%

6406535903503. ✖ 63%, 37%

Question Number : 286 Question Id : 6406531815900 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

What is the expected rate of return of the minimum variance portfolio?

Options :

6406535903504. ✔ 11.07%

6406535903505. ✖ 13.87%

6406535903506. ✖ 15.37%